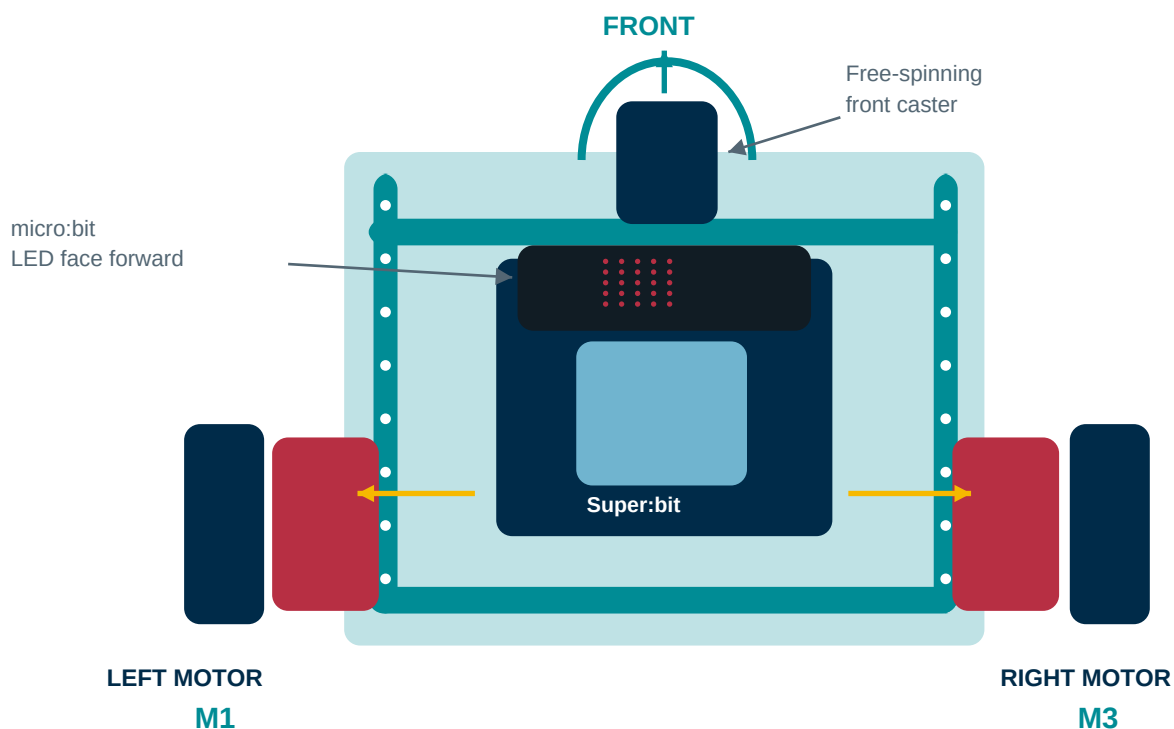




Build a joystick-controlled rover

A complete, illustrated build-and-drive guide for the three-wheel ThinkerLab rover.

Two rear drive wheels. One free-spinning front support. One iPhone.



Layout illustration - not a parts-scale drawing. View from above; front is toward the LED face.

Your route from box to moving robot

BUILD	INSTALL	DRIVE
Parts + chassis Pages 2-4	Firmware + iPhone app Pages 5-6	Connect + your control mode Pages 7-10

Before you start

Use a Mac with Xcode and a micro:bit V2. This is a source-installed iPhone app, not an App Store download. Start with the wheels lifted.



01 / Gather your parts

A stable rover starts with the correct motors, connectors and a supported phone.

Quantity	What you need
1	Yahboom Building:bit Super Kit / Super:bit board, plus a micro:bit V2. Check which pieces your kit edition includes.
2 + 2	Two red DC drive motors and two matching wheels. Use equal wheel diameters.
1	A free-spinning front wheel in a swivel fork (caster), or a ball caster with a compatible bracket. Add one if your kit lacks it.
As needed	Rigid beams/base plate, cross-braces, axles/bushes and compatible pins or fasteners. No soldering is required for the standard connectors.
1 set	The correct Super:bit battery and its approved charging arrangement; data cables for micro:bit and iPhone; a stable phone stand.
1 each	Mac, Apple Account and iPhone. Internet is needed for software downloads/signing.

Software and compatibility

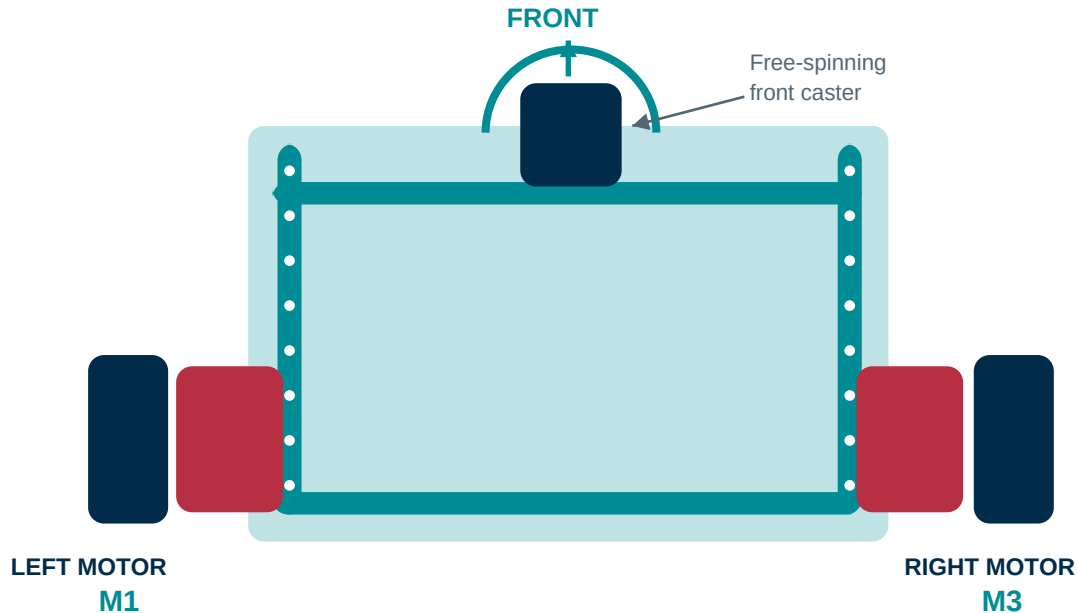
These guides target app **1.7.2 (build 15)**, built with Xcode 26.5 and tested on iPhone 17 Pro / iOS 26.6. The project targets iOS 17+, but older devices have not been validated. Use Xcode 26 or newer with an SDK/device-support version compatible with your phone.

Joystick needs Bluetooth only. No camera, speech model or Apple Intelligence is needed to drive.



02 / Build a rigid three-wheel chassis

A custom triwheel layout using compatible kit parts; exact brick lengths can vary.



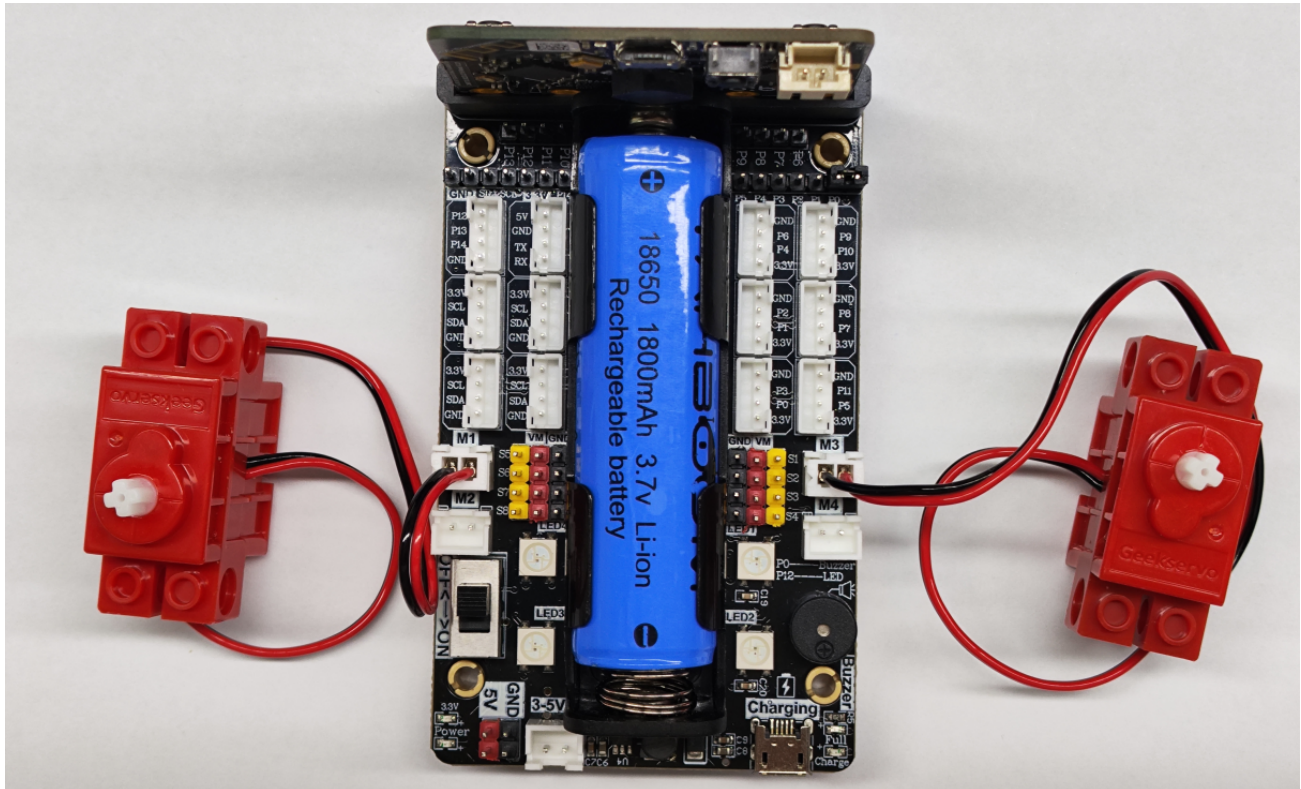
- 1 Make a braced base**
Join two parallel beams with front and rear cross-braces. Add a plate or a second brace so the frame cannot twist when you gently flex opposite corners.
- 2 Fit the rear motors and wheels**
Mount one DC motor on each rear side. Keep shafts level, parallel and facing outward. Fit the two drive wheels; leave a small gap so tyres and hubs do not rub on beams.
- 3 Add the front support**
Centre the caster at the front. Fit its fork/axle loosely enough to roll freely but securely enough not to fall out. A swivel or ball caster makes rotation easier; a fixed front wheel may scrub sideways.

CHECK: all three wheels touch a flat surface; the chassis sits level; rear wheels rotate freely; the caster swivels through a full turn. Adjust bracket height before adding electronics.



03 / Mount, connect and power

Turn the board OFF and disconnect charging/USB power before moving connectors.



Yahboom reference photo: left motor on M1, right motor on M3. Match the printed labels on your board; this is not a photo of the custom chassis. [2]

4

Secure the board and insert micro:bit

Use the Super:bit mounting holes and compatible pins/standoffs. Keep metal away from exposed contacts. Insert the micro:bit fully in the edge connector, oriented as the manufacturer shows; keep its LED face toward your chosen front.

5

Connect the drive motors

Viewed from behind, looking forward: plug the LEFT motor into M1 and RIGHT motor into M3. Use the keyed two-wire plugs without forcing them. Leave M2, M4 and the servo sockets unused. Route leads clear of the wheels.

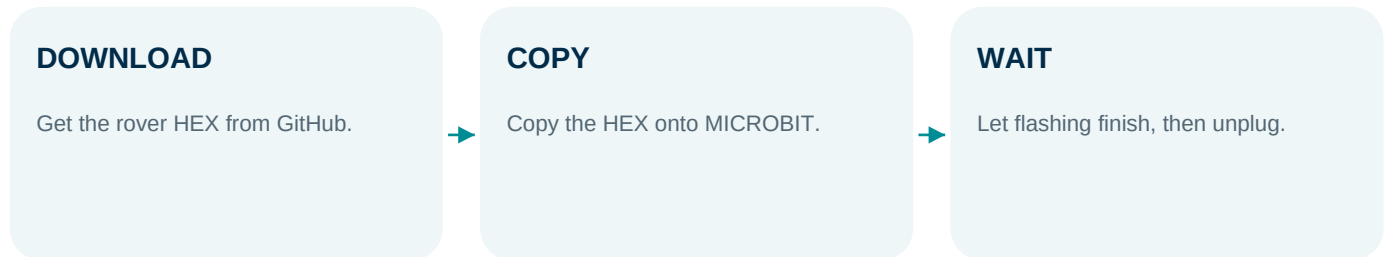
Battery and charging

Use only the battery type/polarity specified for your board revision. The pictured board uses one 3.7 V 18650 cell. Charge through the Super:bit charging port, not the micro:bit USB port; do not drive while charging. Keep its switch OFF until testing. [3]



04 / Put rover firmware on micro:bit

Firmware is the small program that receives Bluetooth commands and controls the motors.



- 1 Download the correct program**

Open [the firmware download](#) and save **microbit-rover.hex**. Or open [GitHub / downloads](#) and use the file download button. This replaces the program already on micro:bit. Do not use the vendor demo firmware or the LED-only example.
- 2 Connect micro:bit to the Mac**

Leave motor power off. Connect a USB data cable to the micro:bit itself. Finder should show a drive named **MICROBIT**. A charging-only cable will not work.
- 3 Copy and wait**

Drag microbit-rover.hex onto MICROBIT. Wait for the transfer and the flashing activity LED to finish. If a FAIL.TXT file appears, read it and retry with a known data cable. The supplied firmware shows a small diamond while disconnected.
- 4 Disconnect the programming cable**

The rover will run wirelessly once its board battery is switched on. A USB cable in the micro:bit may power only the controller; it is not a substitute for the board motor supply.

Optional: build the HEX from source

Install Node.js/npm, download/extract the repository, and run these commands in its folder. The build script uses the MakeCode cloud compiler, so internet is required.

```
bash scripts/build-firmware.sh
```

```
cp -X .build/microbit-rover.hex /Volumes/MICROBIT/
```

The ready-made HEX is supplied for this guide; no Arduino or ESP32 firmware is needed.



05 / Install Microbit Link on iPhone

Use your own Apple Account and signing identity. There is no App Store or TestFlight link.

- 1 Get the current source**

Open [the project on GitHub](#). Choose **Code > Download ZIP**, then extract it on your Mac. Open **MicrobitLink.xcodeproj** from the extracted folder. Use version 1.7.2 or later; the older release still has Enable buttons and no voice screen.
- 2 Set up Xcode signing**

Install a compatible Xcode from Apple and open it once to finish setup. In Xcode Settings > Apple Accounts, sign in. Select the blue project, then the MicrobitLink target > Signing & Capabilities. Enable automatic signing and choose your own Team.
- 3 Choose an app identifier**

Replace org.example.microbitlink with a unique identifier such as **com.yourname.rover**. Keep that identifier for later updates; changing it creates a separate app with separate preferences.
- 4 Trust and prepare the phone**

Connect and unlock iPhone, accept Trust prompts and select it as Xcode's run destination. If requested, enable Settings > Privacy & Security > Developer Mode, restart and confirm. Developer Mode may appear only after pairing with Xcode. [4,5]
- 5 Build, install and open**

Click the Run triangle (Command-R). Wait for a successful build and installation. Follow any developer-trust prompt on iPhone; then open **Microbit Link** and allow Bluetooth. The app screen is titled **Microbit Rover**. [4]

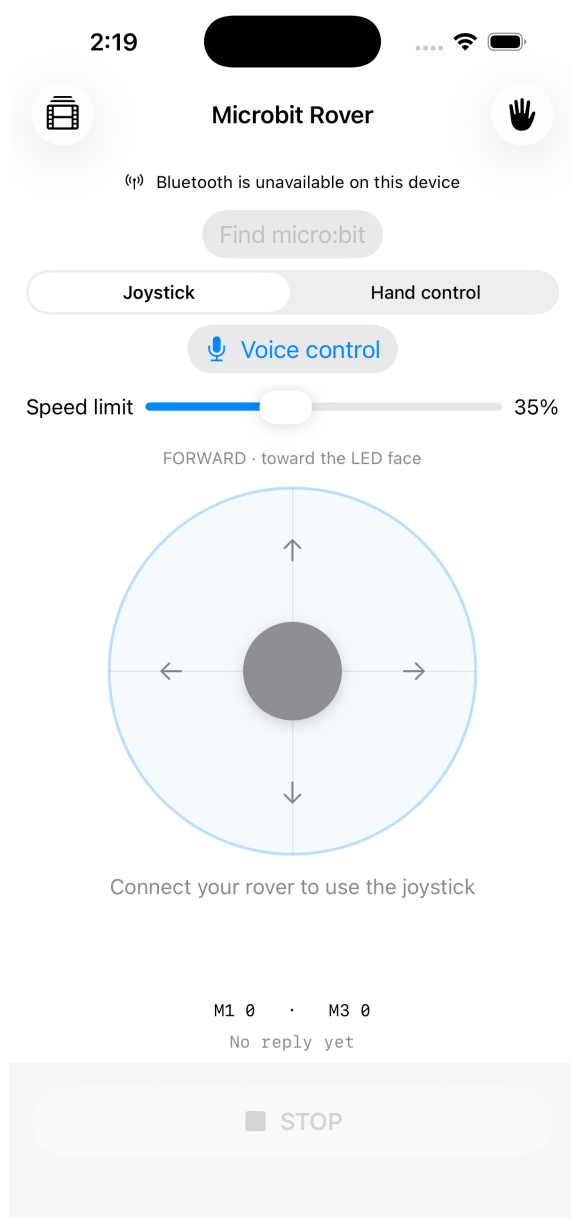
Keep the installation working

A free Personal Team installation expires after seven days; rebuild and reinstall with the same identifier to renew it. If Xcode cannot support your iOS version, update Xcode/macOS as required before retrying. An iPhone alone cannot perform this source-install workflow. [6]



06 / Connect and check wheel direction

Do this once with joystick control before trying gestures or speech.



Actual app UI in the iPhone simulator. Bluetooth is unavailable there; your physical phone shows nearby boards after scanning.

1

Lift the drive wheels

Support the chassis securely so both rear wheels are off the table. Switch on the Super:bit battery supply.

2

Connect inside the app

Tap Find micro:bit, then select your board. Wait for Connected - controls ready. Connect in this app, not the iPhone Bluetooth Settings list.

3

Try a tiny forward touch

Select Joystick and use the default 35% limit. Touch briefly above centre, then release. Both wheels should drive toward the marked front.

4

Correct the mapping

Tap STOP > Wheel setup. Keep "M1 is the right wheel" OFF for M1-left/M3-right. Reverse only a motor that turns backward. Tap Done and test again.

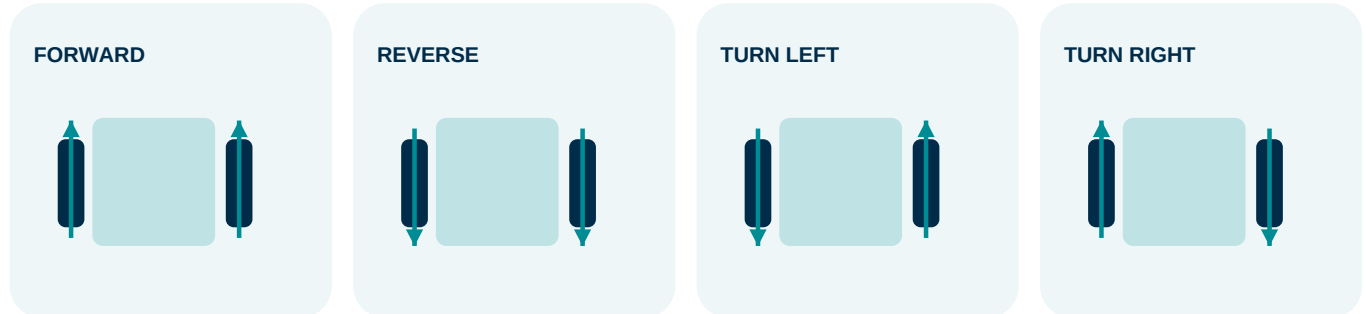
Pass this check before floor driving

Release stops both wheels. Left/right rotate the expected way. STOP and either micro:bit button stop movement. A new touch is needed after Stop. There is no separate Enable step: the app handles ARM when fresh input arrives.



07 / Drive with your thumb

Start on a clear floor at low speed; keep STOP and the board power switch accessible.

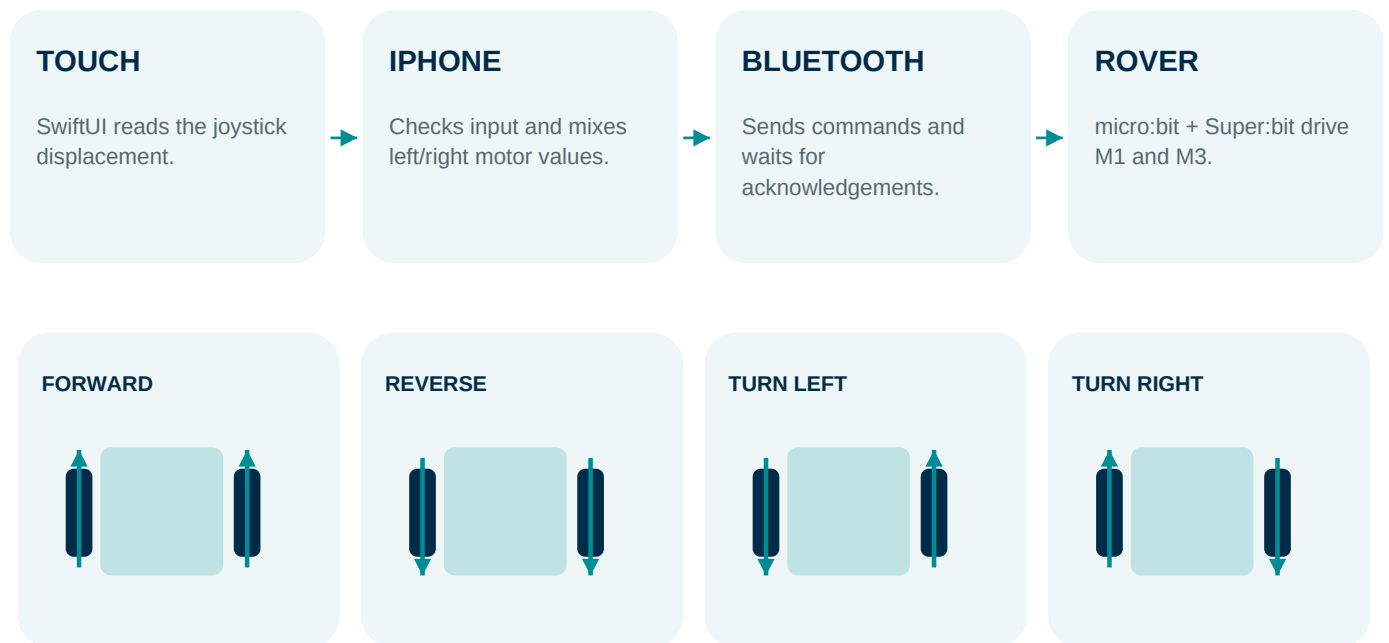


- 1 Touch and hold**
Choose Joystick. Touch near the centre and keep your finger on the screen.
- 2 Move in the direction you want**
Slide up to go forward or down to reverse. Slide sideways at the centre to rotate. Slide diagonally to combine forward/reverse with a turn.
- 3 Choose the amount of movement**
Small displacement gives less output. More displacement gives more output up to the Speed limit. The control has a small centre dead zone.
- 4 Release to stop**
Lift your finger to send zero motor output. The joystick springs back to centre. Tap the red STOP button to cancel and disarm; use a fresh touch to resume.
- 5 Practice before increasing speed**
Try forward for one second, release, reverse briefly, release, then a small turn each way. Stop before changing the speed slider or wheel setup.



08 / Understand the technology

The iPhone interprets your input. The micro:bit runs the motor program.



Differential drive

There is no steering servo. Equal wheel output moves straight. Different outputs make an arc; opposite directions rotate on the spot. The front caster simply follows.

How the app mixes motion

The joystick provides forward and turn values. The app combines them into $\text{left} = \text{forward} + \text{turn}$ and $\text{right} = \text{forward} - \text{turn}$, then applies limits and the saved wheel swaps/reversals.

Why your thumb controls speed

The centre dead zone avoids accidental creep. Outside it, displacement scales the output. The slider selects 20-60% of the driver scale, defaulting to 35%; this is not a measured vehicle speed.

The stop chain

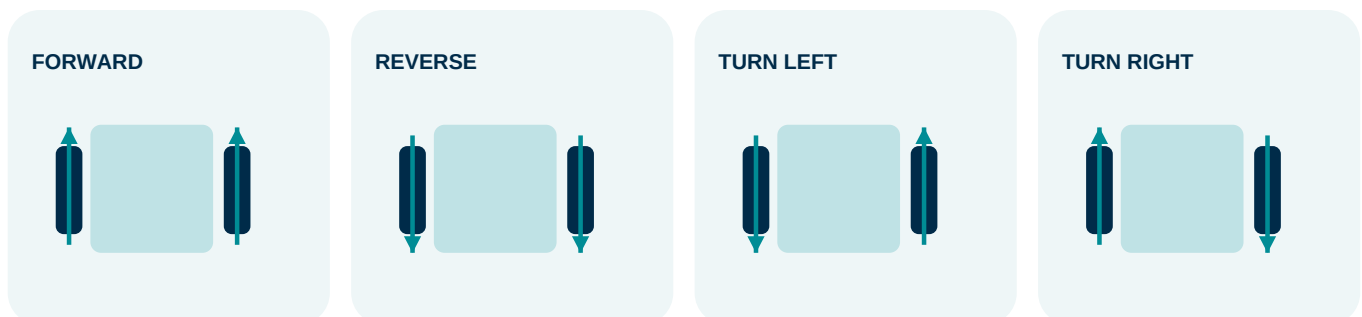
The app sends motor updates at 10 Hz, with one outstanding reply. A reply timeout around 350 ms triggers Stop/disconnect. Firmware separately disarms after 400 ms without a valid drive update. These are software checks; wheels can coast, and they do not detect obstacles.



09 / Practise, tune and explore

Use short, repeatable trials so you can tell what changed.

- 1 Build a release habit**
Mark a short lane with tape. Drive forward for about one second, release, then reverse back. Check that release stops the wheels every time.
- 2 Compare rotation and an arc**
Move sideways from centre to rotate. Now move diagonally forward-right: both motion and turn are combined. Explain why the two wheel outputs differ.
- 3 Tune one setting at a time**
Use STOP before changing speed. Keep wheel reversal consistent with forward direction. A rover that drifts can have unequal friction, battery load or motor performance; there is no encoder feedback to correct it.
- 4 Run a low-speed course**
Use wide turns and soft markers on a level floor. Do not use table edges, stairs or people as obstacles. The rover has no automatic obstacle avoidance.
- 5 Switch to another control mode**
Stop, then choose Hand control or Voice control. The same firmware and wheel settings are reused; there is no need to rebuild the rover or reflash between modes.



Challenge: predict the sign of M1 and M3 before each turn. Then compare your prediction with the app display.



10 / Troubleshoot in the right order

Stop first. Check the visible evidence before changing wiring or settings.

Symptom	Check / next action
No MICROBIT drive	Try a known USB data cable and another port. Connect to micro:bit, not the board charging port. Check FAIL.TXT after a failed flash.
Xcode will not install	Unlock/trust the phone; select the correct device and your own Team/unique bundle ID. Update Xcode if the phone OS is unsupported. Renew an expired Personal Team install.
No rover found	Switch on the board; allow Bluetooth in iPhone Settings; scan in Microbit Link. Disconnect other clients. Reflash the rover HEX if vendor firmware is still installed.
Connected but motors do not run	Check the Super:bit switch, charged correct battery, M1/M3 connectors and wheel clearance. USB in micro:bit alone may not supply the motors. Test joystick first.
Directions are wrong	STOP > Wheel setup. Confirm M1-left/M3-right. Reverse only the motor that is backward. Swap wheels only if physical left/right really are exchanged.
Stops unexpectedly	Read the rover reply. SAFE:BUTTON means a physical stop; SAFE:TIMEOUT means updates stopped. Reply timeout needs reconnecting. Start again with fresh input, not a held old touch.
Turns poorly on the floor	Check caster swivel, rubbing tyres, level chassis and battery. Test the same input with wheels lifted. A stalled motor should not be left powered.
Cannot move after STOP	Lift your finger and make a new touch. Stop intentionally invalidates the old held gesture. A dropped Bluetooth connection still needs reconnecting.

If the problem remains: record the app version, exact status/reply, control mode, motor values, and whether the wheels moved while lifted. Share a wiring photo showing M1/M3 labels and describe one repeatable test.



Keep this page beside the rover

Session checklist, recovery actions and source links.

Every session

1. Check wheels, loose leads and battery. 2. Connect in Microbit Link. 3. Test a short movement and Stop with wheels lifted. 4. Put the rover on a clear floor. 5. Keep the app foreground and watch the rover. 6. Finish with STOP, switch off board power, then store safely.

Know how to stop

Joystick: release. Gesture: open pinch or remove hand. Voice: say Stop, or use the screen if speech is delayed. Red STOP and either micro:bit button cancel movement. If software is unresponsive, switch off board motor power. Nothing here senses a cliff or obstacle.

Downloads and references

[1] [Source, app and firmware](#)

[2] [Yahboom kit and motor wiring](#)

[Yahboom M1/M3 photo source](#)

[3] [Board charging instructions](#)

[4] [Apple: run on a physical device](#)

[5] [Apple: Developer Mode](#)

[6] [Apple: Personal Team and renewal](#)

[ThinkerLab](#)

GitHub: github.com/ibut-bot/iphone-microbit-rover

Kit: yahboom.net/study/buildingbit-super-kit

Brand: thinkerlab.com.au

Edition notes: app 1.7.2 / build 15, source snapshot a71c3e5. Joystick and gesture use were confirmed on the existing rover. Voice forward and ordinary right-turn were confirmed; the latest polite-phrase fix passed automated tests and still needs user confirmation. Triwheel drawings are a proposed custom layout, not a verified kit bill of materials.

Image credits: ThinkerLab logo from its website; motor/board photograph from Yahboom. App screenshots are simulator captures. All labelled chassis, control and data-flow diagrams were created for these guides. Refer to the linked manufacturer instructions for your exact board revision.

